Bar Tack Machine



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bar-tack machine

What it is used for

A bar tack machine sews short, dense reinforcement stitches at stress points on garments and textile products. Use the correct stitch program, thread, needle, and fabric positioning to produce secure tacks while preventing needle injuries and machine damage.

Main parts

- Machine head and sewing needle
- Presser foot and fabric guide
- Needle and bobbin thread system
- Stitch pattern and length controls
- Foot pedal or electronic start control
- Motor, needle guard, and emergency stop, if fitted

Before you start

1. Confirm the required bar-tack pattern, stitch density, fabric type, and reinforcement location.
2. Inspect the needle, presser foot, thread path, bobbin, guards, and power cable for damage.
3. Ensure the needle is correctly fitted, sharp, and suitable for the fabric and thread.
4. Thread the machine and install the bobbin according to the manufacturer's instructions.
5. Select the approved stitch program and verify the settings on a sample piece.
6. Keep the work area clear and wear suitable eye protection where needle breakage or flying fragments are a risk.

How to use

1. Switch off the machine before changing the needle, threading near the needle, or clearing a jam.
2. Select the correct bar-tack program and set the approved stitch length, width, and density.
3. Position the fabric so the reinforcement area lies under the needle and within the designated guide.
4. Lower the presser foot and ensure the fabric is held flat without folds or excessive tension.
5. Keep fingers away from the needle path and activate the machine using the approved control.
6. Observe the first tack for correct placement, stitch formation, and secure thread locking.
7. Stop the machine if the needle breaks, thread tangles, fabric shifts, or abnormal noise occurs.
8. Raise the presser foot and remove the fabric only after the needle has stopped in a safe position.
9. Inspect the tack for skipped stitches, loose threads, incorrect length, and fabric damage.
10. Trim loose threads with suitable scissors while keeping hands clear of the needle.
11. Switch off the machine and clean lint from the needle plate, bobbin area, and accessible guides.

2. Record quality issues and arrange adjustments or maintenance if the stitch pattern is inconsistent.

⚠ Safety

- Only trained users should operate or adjust the bar-tack machine.
- Keep fingers away from the needle and presser foot while the machine is running.
- Never bypass needle guards or other safety devices.
- Switch off and isolate power before replacing a needle, servicing electrical parts, or clearing a jam.
- Do not pull fabric forcibly while the needle is engaged.
- Keep hair, jewellery, and loose clothing away from moving mechanisms.

Care & storage

- Remove lint and thread waste from the needle plate and bobbin area after shutdown.
- Inspect and replace bent, blunt, or damaged needles promptly.
- Check thread tension, bobbin condition, presser foot, and needle alignment regularly.
- Lubricate designated points only as directed by the manufacturer.
- Arrange servicing if stitch density, timing, or machine movement becomes irregular.

Common mistakes to avoid

- Using the wrong stitch program or needle for the fabric.
- Placing fingers too close to the needle during sewing.
- Pulling the fabric while the needle is engaged, causing needle breakage.
- Ignoring skipped stitches, loose thread locking, or repeated thread breakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.